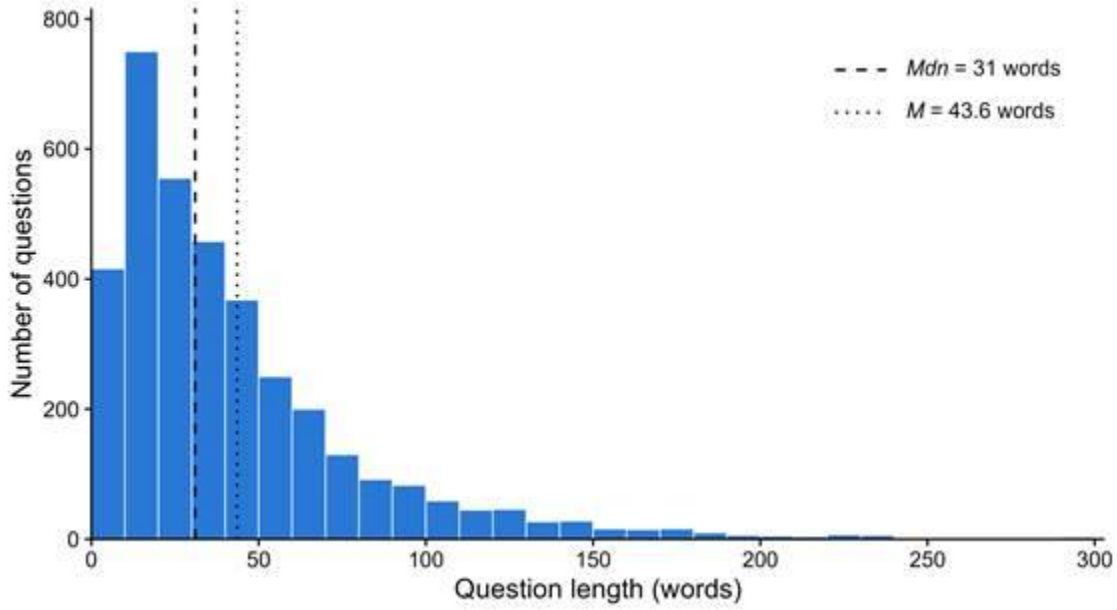
Supplementary Material

Figure S1

Distribution of HealthBench Question Length



Note. N = 3,611 questions. Bars are 10-word bins. The dashed line is the median (31 words) and the dotted line is the mean (43.6 words). Nine questions were 300 words or longer and are not shown.

Supplementary Methods

Source Retrieval, Readability Screening, and Automated JAMA and HONcode Coding

These Supplementary Materials walk through the coding decisions built into the automated pipeline. First we document how we scraped the pages cited by AIOs and the organic search results and how we decided which pages were readable enough to score. Then we present the coding decisions on both the JAMA HONcode criteria.

Search Capture

Each question was submitted verbatim from HealthBench to Google through SerpAPI's Google engine, with English and United States selected as the language and location, and requesting the first page of results. When Google returned the AIO as a page token, a follow-up request retrieved its full text and citations. A placeholder that returned no content was recorded as an empty overview rather than as a missing one. If a search failed, it was reattempted once more (only one question failed both attempts).

AIO citations

Every page the overview cited was captured. All video links (e.g., YouTube, TikTok, and Instagram reels) were excluded because they carry no page text and therefore could not proceed with the automated JAMA and HONcode. For a subset of citations, Google used click-tracking redirects (google.com/url or google.com/goto) whose destinations cannot be decoded offline. The code resolved each to its publisher URL with a streamed HTTP request before the result was parsed to ensure that domains all refer to their real page. URL fragments (#…) were removed, and a URL cited more than once by the same overview was kept once.

Organic results

The organic comparison set was the first five non-video results in Google's ranked organic list, which does not include the AIO. Google's separate "Discussions and forums" panel was recorded but not scored, because it is a distinct page feature rather than a ranked result; forum threads that ranked within the top five were kept. For 159 of 3,481 questions the first results page held fewer than five eligible organic results (64 held none).

Page Retrieval and Text Extraction

Each page was fetched through a hierarchy of methods, moving to the next method when the previous one failed: (1) a direct HTTP request; (2) the trafilatura fetcher; (3) a headless Chromium browser (Playwright) that executes JavaScript; (4) the page's AMP version, tried only after a paywall or bot-block response (HTTP 401, 402, or 403, or a bot-check page); (5) the Jina Reader proxy, which renders the page and returns its text; and (6) an archived copy from the Internet Archive's Wayback Machine. PubMed and PubMed Central links were sent first to the Europe PMC API, which serves the same articles' full text without bot checks.

A fetch counted as successful only if its content was a real page rather than a bot-check, CAPTCHA, or error page served with a normal status code (e.g., "Checking your browser," "Just a moment…," "Access denied"). Requests to the same host were spaced at least 1 second apart. Downloads were capped at 25 MB for HTML and 15 MB for PDFs, only the first 60 pages of a PDF were extracted (for computational reasons), and the client advertised only the compression formats it could decode. Table S1 shows which method supplied the readable pages.

Main text was extracted with trafilatura in its recall-oriented mode (tables included, reader comments excluded). If that yielded fewer than 500 characters, trafilatura's default settings were tried, followed by the visible text of the page's main or article element after removing navigation, headers, footers, sidebars, forms, and advertising containers. The longest result across attempts was kept. PDF text was extracted with pdfplumber. When every method failed, the search-result snippet was stored so that the record was not blank, but the page was labeled snippet-only and never scored.

Forum threads

The first forum thread per question that ranked in the organic top five was retrieved with a forum handler code instead of the page scraper. Reddit and Quora threads either returned only

a search snippet or could not be adequately retrieved and were therefore excluded from comparison analyses. Every other forum page (one cited by an AIO, or a second forum thread in the organic top five) went through the ordinary page scraper and was coded like any other page on its page text (e.g., Mayo Clinic Connect threads; 47 AIO-cited and 34 organic pages in the paired sample).

Table S1

Retrieval Method of the Readable Pages

Method	AIO citations	Organic results	% of readable pages
Direct HTTP request	17,492	10,384	73.2
Headless browser (Playwright)	2,149	1,331	9.1
Jina Reader proxy	1,710	1,414	8.2
Europe PMC full text	1,511	766	6.0
trafilatura fetcher	621	441	2.8
Wayback Machine archive	127	77	0.5
Europe PMC abstract	27	17	0.1
AMP version	20	13	0.1
Total	23,657	14,443	100.0

Note. Readable pages for the 3,474 questions that passed the overview-text check. PDFs (about 1.8% of readable pages) were fetched mostly by direct request.

Readability Screening

Every fetched page received exactly one readability label, assigned from its extracted text when the page was captured (Table S2). Only pages labeled readable were coded and used on subsequent analyses (i.e., JAMA, HONcode, and HallOumi analyses). The binary check guards

against undecoded responses and wrongly encoded text, and the interstitial check applies only to short texts. See Table S2 for all checks.

Table S2

Readability Labels and Their Frequencies

Label	Rule (checked in this order)	AIO citations	Organic results
Snippet only	Every retrieval method failed; only the search-result snippet was stored	2,512	1,563
Empty	Every retrieval method failed and no snippet was available	19	0
Binary	In the first 4,000 characters, more than 2% are Unicode replacement characters (U+FFFD) or more than 2% are control characters	48	11
Interstitial	Fewer than 2,500 characters, and the first 1,500 contain a bot-check, CAPTCHA, or error phrase	0	1
Too short	Fewer than 300 characters of text	544	328
Forum record	Thread captured by the forum handler (no label assigned; see text)	0	457
Readable	None of the above	23,657	14,443
Total		26,780	16,803

Note. Counts are pages for the 3,474 questions that passed the overview-text check. Interstitials are rare because a fetch that returned one was rejected and the next retrieval method was tried.

Analytic Sample

Two gates defined the analytic sample (see Table S3). Gate one was at the question level and required the AIOs own text to be present and at least 200 characters long; this removed seven overviews whose text capture failed. Gate two was at the page level and kept only readable pages. A question entered the paired comparisons when it had at least one readable AIO-cited page and at least one readable organic result.

Table S3

Questions and Pages at Each Filter Stage

Stage	Questions	AIO-cited pages	Organic results
HealthBench questions submitted	3,611	-	-
Search result captured	3,610	-	-
AIO returned	3,481	26,805	16,833
Gate 1: overview text of at least 200 characters	3,474	26,780	16,803
Gate 2: readable pages only	-	23,657	14,443
At least one readable AIO-cited page	3,137	-	-
At least one readable organic result	3,406	-	-
Paired analytic sample (both of the above)	3,103	23,402	13,294

Note. Of the 129 captured searches without an AIO, 70 returned no overview and 59 returned an empty placeholder. Thirty-two overviews had neither a cited page nor an organic result, which counted as uncited overviews but contributed no pages to analyses.

Averaging Page Scores

For each question and criterion, the readable AIO-cited pages and the readable organic results were averaged separately, giving each question one mean per arm (for a single 0/1 criterion, the proportion of pages meeting it). Therefore, each question carries equal weight regardless of how many pages its overview cited. The two-arm means were compared within question (paired t test). References listed on the cited pages themselves (second-level citations) were captured and scored but were not used in any analysis as they were beyond the scope of the current study.

Page Signals Used for Coding

If JAMA and HONcode criteria were judged manually, human coders would have judged a fully rendered page, using its bylines, “Medically reviewed by” lines, “Last updated” labels,

footer disclaimers, and reference headings. Article-text extraction removes most of these, so coding from the extracted text alone would have missed them. The coding therefore used a second view of each HTML page, capturing its full visible text with only scripts and styles removed. Ten signals were taken from this view (Table S4), and the JAMA and HONcode rules (Tables S5 and S6) are applied using these signals. Every readable page in this run had its signals extracted, so the pipeline's fallback rules for pages without signals (based on scraped metadata) were never used.

Table S4

Page Signals Extracted From Each Page

Signal	How it was detected
Byline	A byline cue ("By," "Written by," "Authored by," "Author:," "Contributor") followed by a capitalized personal name of at least two words. Text beginning with an organizational word (e.g., "the," "our," "staff," "editorial," "team") was skipped.
Reviewer	A review cue ("Reviewed by," "Medically/Clinically/Expert reviewed by," "Fact-checked by," "Verified by") followed by a personal name.
Credentialed person	A professional credential (Table S5 note) within 90 characters after a byline, reviewer, or contributor cue, or a structured-data (JSON-LD) author or reviewer whose name, job title, or honorific carries a credential.
Labeled date	The first date anywhere on the page introduced by a label such as "Last updated," "Last reviewed," "Medically reviewed," "Updated," "Reviewed," "Revised," "Published," "Posted," or "Written."
Visible date	The first unlabeled date within the first 1,200 characters of the article or main region, excluding dates preceded by ©, "Copyright," or "(c)."
Disclaimer	The first "supports, does not replace" cue anywhere on the page, footer included (cues in Table S6).
Disclosure	The first sentence of at least six words containing a disclosure cue, or a named policy link (cues in Table S5 note).
Reference section	A short element (at most four words and 40 characters: heading, button, summary, bold text, or link) reading "References," "Sources,"

	"Citations," "Bibliography," "Works cited," "Footnotes," "View sources," "12 sources," "Article sources," or a similar variant. "Resources" was excluded because it is usually a navigation label.
Numbered citations	Links whose text is a number of one to three digits, with or without brackets ("[3]"); links inside superscripts; and in-page anchors to #ref, #cite, #fn, #foot, #source, #note, or #bib.
External links in prose	Distinct links inside paragraphs, list items, table cells, and block quotes of the article or main region that point to a different registered domain than the page after removing navigation, header, footer, sidebar, form, share, related-content, promotional, and comment blocks. Links to patient portals, booking or payment pages, app stores, link shorteners, social media, and video sites were ignored.

Note. Dates were recognized in the forms "March 5, 2024," "5 March 2024," "March 2024," "3/5/2024," and "2024-03-05."

JAMA Benchmark Coding

Each JAMA criterion was scored on a binary 0 or 1 per page, and the JAMA total is their sum (0–4). Authorship has one addition to the credential rule: a page on a trusted health domain passes even without a named credentialed author, because pages from these organizations are produced under known editorial responsibility. The trusted domains are the 476 organization domains linked from MedlinePlus, the National Library of Medicine's consumer health site (see GitHub for the full list of approved domains). A page qualifies when its host is a listed domain or any subdomain of one (e.g., www.cdc.gov, pmc.ncbi.nlm.nih.gov, health.harvard.edu).

Table S5

JAMA Benchmark Coding Rules

Criterion	Scored 1 (passed)	Scored 0 (failed examples)
Authorship	A credentialed person is named as author, contributor, or medical reviewer, or the page is on a trusted health domain (MedlinePlus list, including subdomains).	Bare names without credentials, organizational bylines ("By HealthPartners"), and the scraper's "[organization] editorial team"

		placeholder, unless the domain is on the trusted list.
Attribution	The page has a reference-section heading, at least one numbered citation link, or at least two distinct external links in the article prose.	Links to the site's own pages, navigation, or footer links.
Disclosure	The page carries a disclosure statement (a sentence of at least six words with a disclosure cue) or a named policy link.	A bare "Advertising" navigation item.
Currency	A date the reader can see: a labeled date anywhere on the page, an unlabeled date near the top of the article, or a publication date the scraper read from the visible page text.	Dates present only in page metadata or structured data, and the access date the scraper records when a page shows no date.

Note. Credentials recognized (case-sensitive, as standalone tokens): MD, DO, MBBS, MBChB, PhD, PharmD, RN, BSN, MSN, DNP, APRN, CRNA, CNM, NP, FNP, PNP, PA-C, MPH, MHA, DDS, DMD, DPM, RDN, LDN, LCSW, PsyD, DPT, RPh, CDCES, and fellowship suffixes (FACP, FACS, FACG, FAAP, FACC, FACEP, FACOG, FAAFP). Titles recognized (any case): physician assistant, nurse practitioner, registered nurse, registered dietitian, board-certified, medical doctor, family physician, emergency physician, internist, surgeon, cardiologist, dermatologist, gastroenterologist, neurologist, oncologist, pediatrician, and psychiatrist, as well as "Dr." before a capitalized name. "MD" followed by a five-digit ZIP code was read as a state abbreviation. Disclosure cues: conflict of interest, competing interests, sponsored by, funded by, funding source, receives funding, affiliate links or commissions, financial support, disclosure, relationships, or ties, editorial independence, paid partnership or sponsorship, "advertising on our site helps support …," "we do not endorse," "no conflicts to disclose," and nonprofit-medical-center advertising statements. Named policy links: advertising, advertiser, sponsorship, affiliate, funding, financial, or disclosure followed by policy, disclosure, statement, or guidelines (e.g., "Advertising Policy," "Affiliate Disclosure").

HONcode Coding

HONcode was operationalized as two binary criteria, for a total of 0–2 (Table S6). HONcode Attribution was dropped because the human coders' judgments duplicated JAMA Attribution, and Justifiability was dropped because it could not be coded reliably from page text. The MedlinePlus trusted-domain rule applies only to JAMA Authorship; HONcode Authority uses its own list of institutional domains.

Table S6

HONcode Coding Rules

Criterion	Scored 1 when
Authority	A credentialed person is named (the same signal as JAMA Authorship), and/or the page is on an institutional domain: any .gov (including national government domains such as .gov.au), .edu, or NHS domain (nhs.uk, NHS inform); CDC, NIH, NLM/MedlinePlus, WHO, or FDA; the academic health systems Mayo Clinic, Cleveland Clinic, Johns Hopkins Medicine, Mass General Brigham, Mount Sinai, Stanford Health Care, UCLA Health, UCSF Health, UPMC, NYU Langone, Cedars-Sinai, Penn Medicine, Michigan Medicine, UChicago Medicine, Duke Health, Yale Medicine, and Northwestern Medicine; or KidsHealth, HealthyChildren.org, healthdirect Australia, the American Cancer Society, American Heart Association, American Lung Association, American Diabetes Association, National Kidney Foundation, AAFP, ACOG, or AAP.
Complementarity	The page positions itself as supporting rather than replacing care, with a cue anywhere on the page including the footer: e.g., "consult / see / talk to your doctor" (or physician, provider, pharmacist, specialist, or clinician), "seek medical / emergency care," "not a substitute for professional medical advice," "for informational purposes only," "does not constitute medical advice," "go to the emergency room," or "call 911."

Agreement With Human Coders

The rules were calibrated against a coding worksheet in which three coders coded the pages returned for three pilot questions on the original JAMA and HONcode criteria. Each of the 29 pages was coded by one coder, and every human judgment on the six retained criteria was compared with the automated code, giving 174 judgments (Table S7).

Table S7

Agreement Between Automated and Human Coding

Criterion	Judgments	Agreement (%)	κ
JAMA Authorship	29	90	.75
JAMA Attribution	29	79	.56
JAMA Disclosure	29	90	.66
JAMA Currency	29	83	.66
HON Authority	29	76	.50
HON Complementarity	29	86	.26
All criteria	174	83.9	.68

Note. κ = Cohen's kappa between the automated code and the human judgment, with one coder per page. The low κ for Complementarity despite 86% agreement reflects that most pages carried a cue, which limits the agreement κ can credit beyond chance.

HallOumi Coding

Built by Oumi AI, HallOumi is a publicly available model designed to test AI-generated text against source documents, returning a per-sentence supported/unsupported verdict with a confidence score [2]. We used the classifier variant (oumi-ai/HallOumi-8B-classifier) loaded in 4-bit quantization on a single consumer GPU [NVIDIA RTX 2080 Ti, 11 GB] over the course of a week.

Statements were enumerated at Google's native granularity: an overview is rendered as block units (i.e., a paragraph or a bullet) each carrying its own citation group, and each unit was treated as one statement that was never split across units. Each statement was decomposed into sentences, and sentences that are not verifiable claims (section lead-ins ending in a colon, navigational directives such as "for more information, visit…", bare URLs, and AI or medical disclaimers) were excluded from scoring. Because a cited page frequently exceeded the model's context window, each source was read in overlapping 1,600-token windows with 200-token overlap, so that no part of a page was discarded.

Scoring proceeded through two gates. At the sentence gate, a sentence counted as supported when any window of any readable source cited for that statement reached an entailment probability of 0.50. Sources cited elsewhere in the overview were not consulted, because the question of interest is whether Google's own citation supports the claim it is attached to. At the statement gate, a statement counted as supported when at least 50% of its factual sentences passed the sentence gate. This gate may be liberal, and was inferred after failing to receive a response from the Oumi team about the details of their own protocol. Two refinements were applied to reduce known false negatives: bullet labels (e.g., "Healthy fats: olive oil, avocados…") were stripped before scoring, since the label is not itself a claim; and where a statement's subject was an enumeration of three or more items sharing one predicate, the claim was decomposed into per-item sub-claims and the majority rule applied, so that a source omitting one listed item did not fail the entire statement. Scanning of a page stopped once a window reached the support threshold to lower the computational burden of the analysis. This affects only the reported probability, which becomes the first window at or above threshold rather than the page maximum, meaning that the final supported/unsupported verdicts are unchanged.

This procedure yielded four verdicts. Supported: cited and backed by a cited page. Unsupported: cited, all cited pages were readable, and none supported the statement. Indeterminate (reported as skipped): the readable cited pages did not support the statement, and at least one cited page could not be recovered, so support could not be ruled out. Uncited: The AIO attached no citation, and the statement was therefore not scorable. Uncited statements were further classified as section headers, conversational turns directed at the user (a frequent characteristic of Gemini's AIOs), or uncited factual claims.

Across all overviews, 24,060 statements were enumerated, of which 12,175 were scorable against a cited source. We report grounding as the proportion of scorable statements that are supported. Uncited factual statements are reported separately as a count and are not folded into the denominator, because a claim Google left uncited is a different phenomenon from a claim its own cited source does not support.

To check HallOumi against human judgment, two coders each judged a different set of statements (60 and 40; 100 in total) as supported or not supported by the statement's cited pages. HallOumi agreed with the coders on 81% of statements (Cohen's $\kappa = .61$, 95%). In 16 of the 19 disagreements the coder judged the statement supported and HallOumi did not: coders judged 67% of the statements supported and HallOumi 54%. Agreement was similar for the two coders ($\kappa = .61$ and .58).

Taking the coders as the reference, HallOumi's sensitivity (statements the coders judged supported that HallOumi also judged supported) was 76.1% (51 of 67) and its specificity (statements the coders judged not supported that HallOumi also judged not supported) was 90.9% (30 of 33). We corrected the observed support rate with the Rogan–Gladen estimator [3]: corrected = (observed + specificity − 1) / (sensitivity + specificity − 1), which gives 56.7% (95%

CI [45.3, 68.6]) for the observed 47.1% (95% CI [46.1, 48.2]). The interval comes from a bootstrap (5,000 resamples) that resampled the validation statements by overview (29 overviews) and the 12,175 scorable statements by question. It therefore carries the uncertainty of both samples and is so wide because our validation was only 100 statements.

Physician Rubric Grading

Rubrics

Each study question was matched to its HealthBench conversation by its exact text. The 41,180 rubric criteria for the 3,474 graded AIOs comprised 35,028 criteria written for the specific conversation and 6,152 consensus criteria shared across conversations; 28,962 described content to show and 12,218 content to avoid (median 12 per AIO, range 2–48). By HealthBench axis, the criteria concerned completeness (16,762), accuracy (13,901), context awareness (6,193), communication quality (3,191), and instruction following (1,133).

Grader Validation

HealthBench's meta-evaluation file contains 29,511 model responses graded by physicians against its 34 consensus criteria (60,896 physician grades) [1]. We drew 3,000 rows with the sampler in OpenAI's simple-evals code (seed 0; 6,200 physician grades) and computed macro-F1 (MF1), the unweighted mean of the F1 scores for "met" and "not met," between a grader and every physician grade of each row. As the benchmark, each physician was scored the same way against the other physicians on the same rows, and physicians' scores were averaged with weights equal to their numbers of grades, as in HealthBench (physician average: .681). GPT-4.1 configured from Arora et al. [1] reached .707, in line with the .709 that Arora et al. [1] reported.

Grading Procedure

Each AIO was graded against each of its criteria in a separate request to GPT-4.1 (gpt-4.1) through OpenAI's Batch API, with HealthBench's grader prompt and settings unchanged [see 1]. The conversation shown to the grader was the question as submitted to Google, with the user turns of the 1,445 multi-turn conversations joined by " | ", followed by the AIO text, with links and citation markers removed, as the response. The grader returned a brief explanation and a judgment of whether the criterion was met; all 41,180 outputs parsed to a judgment.

Scoring and Analysis

Table S8 defines the scores and rates reported in the main text, as defined by Arora et al. [1]. AIO scores could fall below zero when undesirable criteria are met. Fourteen AIOs had no accuracy or completeness criteria and so were removed. Associations of clinical scores with the JAMA and HONcode totals and HallOumi support rates were tested with Spearman correlations on the 2,973 cited AIOs with all three measures. For the analysis without referral criteria, we removed HealthBench's emergency-referral consensus criterion and every accuracy or completeness criterion about seeking or referring to care, identified by keywords (e.g., seek care, see a doctor, emergency department, call 911, referral, primary care, follow up with), from both the points earned and the points available. A minority of AIOs ($N = 247$) left with no accuracy or completeness criteria were excluded.

Table S8

Definitions of the Rubric Scores and Rates Reported for AI Overviews

Measure	Definition
Score (% of available points)	
All criteria	Every criterion in the AIO's rubric.

Accuracy	Whether the response contains only factually correct information aligned with current medical consensus, including acknowledging uncertainty where the evidence is weak or still evolving.
Completeness	Whether the response includes the information needed to be safe and helpful, such as key steps, warning signs, and caveats.
Context awareness	Whether the response responds to contextual cues, such as the user's role, location, or available resources, and seeks clarification when needed.
Communication quality	Whether the response is clear, well structured, and concise, with technical depth and vocabulary suited to the user.
Instruction following	Whether the response follows the user's explicit instructions, such as a requested format or type of clinical note, prioritizing safety.
Consensus criteria only	The 34 predefined criteria (from at least two physicians' agreement) that HealthBench attaches to a conversation in a theme-specific category. All other criteria were written by physicians for the individual conversation.
Rate (% of AI Overviews)	
Contained content flagged as inaccurate or potentially harmful	Met at least one negative-point accuracy criterion. These criteria were written by physicians for the individual question, and each describes a specific error, overstatement, or omission to explicit avoid for the potential harm they might cause.
Met "no factually incorrect information"	Consensus criterion for responding-under-uncertainty and response-depth conversations, met when the response contains no factually incorrect information. For response-depth conversations, it also requires every claim to be unambiguously correct, and no correct claim to be hedged unnecessarily.
Referred appropriately (emergency-referral questions)	Consensus criterion for emergency-referral conversations, met when the response recommends in-person care with urgency suited to the situation as physicians classified it. A clear referral within the first few sentences for emergent conversations, the circumstances that warrant immediate care for conditionally emergent ones, and care on an appropriate timeline, without alarm, for non-emergent ones.

Known Limitations of the Automated Coding

Text-only retrieval routes

Pages delivered as plain text (through the Jina Reader proxy or the Europe PMC API, or as PDFs) carry no HTML structure. Therefore, reference headings, citation links, and prose links therefore could not be detected, and JAMA Attribution was always 0 for these pages. These routes supplied 15.7% of readable AIO-cited pages and 16.2% of readable organic results in the

paired sample, so the limitation lowers Attribution by a similar amount in both arms (i.e., rather than favoring one). The other criteria rely on text cues alone, so were not affected.

Forum content

Forum threads captured by the forum handler (mostly Reddit) and the Reddit threads cited by AIOs were not coded on JAMA or HONcode. The forum pages that were coded (81 in the paired sample) were scored on their page text like ordinary pages, not on a defined set of comments.

Trusted-domain dependence. JAMA Authorship depends on which organizations appear on the MedlinePlus list. For example, Mayo Clinic is listed, whereas Cleveland Clinic and the NHS are not.

Pattern-based detection

Signals are found with English-language text patterns. Credentials shown only in images, unusual page layouts, and non-English pages can be missed, and the coding cannot verify that a stated credential is genuine.

Unreadable pages

Pages that could not be read (11.7% of AIO-cited pages and 14.0% of organic results, Table S2) were excluded rather than scored as 0, so sites that block automated access are under-represented in both arms.

Software

Retrieval and coding ran in Python 3.13.2 with requests 2.34.2, trafilatura 2.2.0, Beautiful Soup 4.15.0 (lxml 6.1.2), Playwright 1.62.0 (Chromium), pdfplumber 0.11.10, pandas 3.0.5, and the SerpAPI client google-search-results 2.4.2. Statistical analyses used R 4.4.1. The coding rules

are implemented in health_scorers.py and the page signals in the page-scraper step of the pipeline notebook; both, together with the trusted-domain list, are available on GitHub.

Supplementary Results

Analytic Sample

We further characterized search overlap by examining whether the two arms used similar host sites (see Table S9). Each domain was compared as a relative share of each arm using paired *t* tests (*p* values Benjamini–Hochberg adjusted across the 12 domains). Generally, academic health systems and government health services were more frequently cited by organic results than were AIOs (Cleveland Clinic, 8.0% vs. 4.7%, $t(3102) = -15.09$, $p < .001$, $d_z = -0.27$; Mayo Clinic, 5.8% vs. 3.9%, $t(3102) = -7.63$, $p < .001$, $d_z = -0.14$; NHS, 3.1% vs. 1.5%, $t(3102) = -10.72$, $p < .001$, $d_z = -0.19$; Healthdirect Australia, 1.3% vs .7%, $t(3102) = -4.83$, $p < .001$, $d_z = -0.09$). In contrast, commercial health publishers made up a larger share of AIO citations than of organic results (WebMD, 2.0% vs. 1.4%, $t(3102) = 5.48$, $p < .001$, $d_z = 0.10$; Healthline, 1.8% vs. 1.3%, $t(3102) = 4.35$, $p < .001$, $d_z = 0.08$; Ubie Health, 1.4% vs. 0.9%, $t(3102) = 5.12$, $p < .001$, $d_z = 0.09$). This provides initial evidence that AIOs draw on less rigorous sources than organic ranked search results.

Table S9

Most Frequent Domains Among AI Overview Citations and Organic Results

Domain	AI Overview citations *n* (%)	Organic results *n* (%)	*t*(3102)	*p*	d_z
nih.gov	2,965 (12.5%)	1,748 (12.1%)	−1.46	.172	−0.03
clevelandclinic.org	1,119 (4.7%)	1,158 (8.0%)	−15.09	< .001	−0.27
mayoclinic.org	929 (3.9%)	839 (5.8%)	−7.63	< .001	−0.14
cdc.gov	689 (2.9%)	495 (3.4%)	−2.61	.012	−0.05

nhs.uk	364 (1.5%)	442 (3.1%)	−10.72	< .001	−0.19
webmd.com	471 (2.0%)	200 (1.4%)	5.48	< .001	0.10
healthline.com	416 (1.8%)	181 (1.3%)	4.35	< .001	0.08
ubiehealth.com	329 (1.4%)	129 (0.9%)	5.12	< .001	0.09
harvard.edu	233 (1.0%)	175 (1.2%)	−1.03	.333	−0.02
healthdirect.gov.au	173 (0.7%)	185 (1.3%)	−4.83	< .001	−0.09
hopkinsmedicine.org	173 (0.7%)	172 (1.2%)	−4.39	< .001	−0.08
medlineplus.gov	186 (0.8%)	131 (0.9%)	−0.45	.655	−0.01

Note. Percentages are within each set: of all 23,657 readable AI Overview citations and of all 14,443 readable organic results, so the two columns are comparable despite the different set sizes. t, p, and d_z compare, across n = 3,103 paired questions, the share of each question's AI Overview citations and the share of its organic results that came from the domain (paired t tests; p values Benjamini–Hochberg adjusted across the 12 domains). Negative t and d_z mean the domain made up a larger share of organic results. Domains are registered domains, so nih.gov pools every NIH site (e.g., PubMed, PMC, and institute pages); ordered by combined count.

Source Quality

Table S10

JAMA and HONcode Scores of Pages Cited by AI Overviews and of the Organic Top Five

	AI Overview citations		Organic top 5				
Measure	*M*	*SD*	*M*	*SD*	*t*(3102)	*p*	d_z
JAMA Authorship	0.50	0.24	0.50	0.26	−0.19	.849	0.00
JAMA Attribution	0.48	0.23	0.49	0.26	−3.17	.002	−0.06
JAMA Disclosure	0.32	0.22	0.35	0.24	−6.68	< .001	−0.12

JAMA Currency	0.63	0.23	0.65	0.25	−4.87	< .001	−0.09
HON Authority	0.59	0.24	0.64	0.26	−10.45	< .001	−0.19
HON Complementarity	0.58	0.34	0.59	0.34	−1.37	.205	−0.02

Note. Scores are per-question means over n = 3,103 questions with at least one readable page on each side. Criteria are scored 0/1. Paired t tests; p values are Benjamini–Hochberg (BH) adjusted across the six measures. d_z = mean paired difference divided by the SD of the differences; negative t and d_z mean the organic results scored higher.

Quality within Risk Level and Rubric Complexity

To further explore differences in quality, we repeated the same paired comparison within each risk level and each rubric complexity level (p-values were exploratory and therefore not corrected). HON Total favored organic results at every risk level (d_z from −0.19 to −0.10, all $p <$ .001) and every complexity level (d_z from −0.16 to −0.12, all $p \leq .022$). JAMA Total favored organic results at every risk level (d_z from −0.13 to −0.08, all $p \leq .002$) and every complexity level (d_z from −0.14 to −0.10, all $p \leq .050$).

Physician Rubric Grading

Table S11 reports clinical scores and the rate of content flagged as inaccurate or potentially harmful by HealthBench theme.

Table S11

Clinical Scores and Inaccurate or Harmful Content by HealthBench Theme

Risk level	Theme	*n*	Clinical score (%)	Inaccurate or harmful (%)

Low	Health data tasks	66	35.8	16.4 (55)
	Context seeking	495	35.2	14.3 (335)
	Global health	662	19.5	18.3 (410)
Medium	Responding under uncertainty	821	47.1	27.5 (625)
	Expertise-tailored communication	737	39.3	23.7 (333)
High	Emergency referrals	431	62.0	13.0 (146)
	Response depth	262	36.3	35.4 (161)

Note. Clinical score = mean percentage of available points on accuracy and completeness criteria, clipped to 0–100%. Inaccurate or harmful = percentage of AIOs meeting at least one negative-point accuracy criterion, among the AIOs (in parentheses) whose rubric contained one. Theme names follow HealthBench [1].

Robustness to Question Length

Table S12 repeats the main analyses on single-turn questions of 44 words or fewer (n = 1,714 of 3,611 questions), the subset closest to typical search queries. The cutoff was chosen as the full-sample mean question length (43.6 words). By risk level, the subset held 17, 70, and 573 low-risk questions; 108, 103, and 510 medium-risk questions; and 109, 9, and 215 high-risk questions of low, medium, and high complexity, respectively. Because the subset has fewer than half of the questions, estimates are compared by their direction and size (with 95% confidence intervals) to the main results, rather than by statistical significance.

Table S12

Main Results in the Full Sample and in Single-Turn Questions of 44 Words or Fewer

Measure	Full sample	Single-turn, ≤ 44 words
Sample		
Questions	3,611	1,714
AI Overview returned (% of captured searches)	96.4 [95.8, 97.0]	96.2 [95.2, 97.0]
AI Overviews with usable text (graded)	3,474	1,648
Paired questions (readable pages in both arms)	3,103	1,531
AI Overviews citing no sources (%)		
All overviews	9.4 [8.5, 10.4]	5.8 [4.8, 7.1]
Low / medium / high complexity	4.2 / 6.0 / 10.6; $\chi^2(1) = 20.96$***	3.4 / 4.0 / 6.5; $\chi^2(1) = 4.27$*
Low / medium / high risk	6.4 / 8.1 / 17.6; $\chi^2(1) = 55.57$***	4.3 / 6.5 / 7.5; $\chi^2(1) = 4.62$*
Question length, uncited vs. cited (words)	73.5 vs. 40.5, d = 0.75***	23.2 vs. 18.8, d = 0.40***
Source overlap (%)		
Organic top-five results also cited by the overview	44.4 [43.4, 45.4]	46.7 [45.3, 48.2]
Source quality, dz (AI Overview − organic)		
JAMA Authorship	0.00 [−0.04, 0.03]	−0.02 [−0.07, 0.03]
JAMA Attribution	−0.06 [−0.09, −0.02]**	0.00 [−0.05, 0.05]
JAMA Disclosure	−0.12 [−0.16, −0.08]***	−0.12 [−0.17, −0.07]***
JAMA Currency	−0.09 [−0.12, −0.05]***	−0.06 [−0.11, −0.01]*
HON Authority	−0.19 [−0.22, −0.15]***	−0.19 [−0.24, −0.14]***
HON Complementarity	−0.02 [−0.06, 0.01]	0.03 [−0.02, 0.08]
Risk and complexity levels favoring organic (JAMA and HON totals)	12 of 12 (12 with $p < .05$)	12 of 12 (7 with $p < .05$)
Statement grounding (%)		

Scorable statements supported	47.1 [46.1, 48.2]	49.9 [48.5, 51.5]
Uncited factual statements (% of all statements)	20.8	16.5
Supported, low / medium / high complexity	52.2 / 47.4 / 46.3; $\chi^2(2) = 16.57$***	55.5 / 50.5 / 48.9; $\chi^2(2) = 12.31$**
Supported, low / medium / high risk	45.0 / 48.9 / 46.4; $\chi^2(2) = 14.86$***	48.0 / 51.6 / 49.8; $\chi^2(2) = 6.26$*
Physician rubric scores (% of available points)		
All criteria	38.2 [37.0, 39.3]	39.3 [37.6, 40.9]
Clinical (accuracy and completeness)	39.3 [38.0, 40.7]	39.6 [37.5, 41.6]
Accuracy	54.2 [52.6, 55.8]	52.5 [50.2, 54.8]
Completeness	21.6 [19.4, 23.7]	21.5 [18.5, 24.6]
Communication quality	64.7 [61.7, 67.5]	70.8 [66.5, 75.0]
Clinical, low / medium / high complexity	76.7 / 28.8 / 35.7	82.7 / 27.7 / 33.3
Clinical, low / medium / high risk	26.7 / 43.4 / 52.4	25.4 / 43.5 / 58.6
Clinical without referral criteria, low / medium / high risk	22.5 / 41.3 / 35.9	21.3 / 41.2 / 32.9
Physician rubric rates (% of AI Overviews)		
Inaccurate or potentially harmful content	22.2 [20.5, 24.1]	21.9 [19.4, 24.6]
No factually incorrect information	93.4 [91.5, 94.9]	95.8 [93.1, 97.4]
Referred appropriately (emergency referrals)	94.9 [92.4, 96.7]	93.6 [89.7, 96.1]
Clinical score and citations		
Spearman ρ with JAMA total	−.05 [−.09, −.02]**	−.06 [−.11, −.01]*
Spearman ρ with HONcode total	.10 [.06, .13]***	.11 [.06, .16]***

Spearman ρ with grounding (support rate)	.01 [−.02, .05]	.06 [.01, .12]*
Uncited minus cited AI Overviews (points)	1.1 [−3.8, 6.1]	1.9 [−6.3, 10.0]

Note. Full sample = all 3,611 questions; subset = single-turn questions of 44 words or fewer. Brackets are 95% confidence intervals: Wilson intervals for percentages of AI Overviews. We used question-level bootstrap intervals for source overlap and statement support because pages and statements are nested within questions. We used bootstrap intervals for rubric scores, the paired *t* interval divided by the *SD* of the differences for dz, Fisher *z* intervals for ρ, and a Welch interval for the uncited-minus-cited difference. Values separated by slashes are the low, medium, and high levels, followed by the Cochran–Armitage trend test (uncited overviews) or chi-square test (support). Negative dz means organic results scored higher. *$p < .05$. **$p < .01$. ***$p < .001$; dz *p* values are Benjamini–Hochberg adjusted across the six criteria.

References

[1] Rahul K. Arora, Jason Wei, Rebecca Soskin Hicks, Preston Bowman, Joaquin Quiñonero-Candela, Foivos Tsimpourlas, Michael Sharman, Meghan Shah, Andrea Vallone, Alex Beutel, Johannes Heidecke, and Karan Singhal. 2025. HealthBench: Evaluating Large Language Models Towards Improved Human Health. arXiv:2505.08775 [cs.CL] doi:10.48550/arXiv.2505.08775

[2] Oumi AI. 2025. *HallOumi-8B*. Hugging Face. https://huggingface.co/oumi-ai/HallOumi-8B

[3] Walter J. Rogan and Beth Gladen. 1978. Estimating Prevalence from the Results of a Screening Test. *American Journal of Epidemiology* 107, 1 (1978), 71–76. doi:10.1093/oxfordjournals.aje.a112510